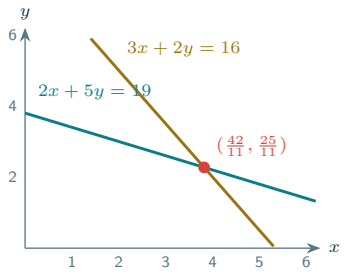
ALGEBRA ON ONE PAGE

Linear simultaneous equations and the LCM

Each equation is a line and the solution is where they cross; the LCM matches a coefficient and the GCD shrinks the numbers first

1 / THE RULE AND THE PICTURE

To eliminate x from $ax + by = c$ and $dx + ey = f$, scale both equations to the coefficient $\mathrm{LCM}(a, d)$



Multiplying an equation by a constant leaves its line unchanged, so scaling to match coefficients and then subtracting only changes how the two lines are written. The multipliers are $\mathrm{LCM}(a, d)/a$ for the first equation and $\mathrm{LCM}(a, d)/d$ for the second:

$$\frac{\mathrm{LCM}(a,d)}{a}\,(1) - \frac{\mathrm{LCM}(a,d)}{d}\,(2) \quad \text{leaves an equation in } y \text{ alone}$$

The same recipe on y uses $\mathrm{LCM}(b, e)$, and a variable with opposite signs on its coefficients is eliminated by adding the scaled equations.

2 / A WORKED EXAMPLE / $3x + 2y = 16$ AND $2x + 5y = 19$

$\mathrm{LCM}(3, 2) = 6$ removes x, and $\mathrm{LCM}(2, 5) = 10$ removes y

	eliminate x	eliminate y
multipliers	$(1) \times 2,\ (2) \times 3$	$(1) \times 5,\ (2) \times 2$
matched equations	$6x + 4y = 32$ and $6x + 15y = 57$	$15x + 10y = 80$ and $4x + 10y = 38$
subtract	$11y = 25$	$11x = 42$
result	$y = \frac{25}{11}$	$x = \frac{42}{11}$

Each elimination gives one coordinate, and substituting back confirms the pair: $3 \cdot \frac{42}{11} + 2 \cdot \frac{25}{11} = \frac{176}{11} = 16$ and $2 \cdot \frac{42}{11} + 5 \cdot \frac{25}{11} = \frac{209}{11} = 19$. The LCM keeps the numbers small when the coefficients share a factor, as in $4x + 3y = 10$ and $6x + 5y = 16$, where $\mathrm{LCM}(4, 6) = 12$ (multipliers 3 and 2) replaces the product 24 and gives $y = 2$, $x = 1$.

3 / SIMPLIFY FIRST / THE GCD OF EACH EQUATION

Dividing an equation by the GCD of its coefficients shrinks the LCM you need

$6x + 9y = 15$	$\mathrm{GCD} = 3$, so $2x + 3y = 5$
$4x + 2y = 10$	$\mathrm{GCD} = 2$, so $2x + y = 5$
subtract	$2y = 0$, so $y = 0$ and $x = \frac{5}{2}$

The x-coefficients of 6 and 4 would have needed $\mathrm{LCM} = 12$, whereas after simplifying they are both 2 and the equations subtract at once.

For $4x+6y = 12$ and $3x+2y = 8$ the first equation divides by 2 to give $2x+3y = 6$, and $\mathrm{LCM}(2, 3) = 6$ gives $6x+9y = 18$ and $6x+4y = 16$, so $5y = 2$ and $y = \frac{2}{5}$. Substituting into $3x + 2y = 8$ gives $3x = 7.2$, hence

$$x = 2.4 = \tfrac{12}{5}, \qquad y = 0.4 = \tfrac{2}{5}$$

4 / WHEN ELIMINATION FAILS / PARALLEL AND COINCIDENT LINES

The number $ae - bd$ decides between one crossing and none or infinitely many

For $ax + by = c$ and $dx + ey = f$ the solution is $x = \dfrac{ce - bf}{ae - bd}$ and $y = \dfrac{af - cd}{ae - bd}$, so a unique crossing needs $ae - bd \neq 0$, and the example above has $ae - bd = 15 - 4 = 11$, which is the 11 in $11y = 25$. When $ae - bd = 0$ the lines share a slope, and they are parallel with no solution if the constants disagree, or the same line with infinitely many solutions if they agree; simplifying $6x + 4y = 20$ and $9x + 6y = 30$ by their GCDs of 2 and 3 gives $3x + 2y = 10$ twice, which is the second case.

5 / PRACTICE / ANSWERS TO CHECK AGAINST THE GRAPH

Four systems, three unique crossings and one repeated line

system	key step	solution	check
$2x + y = 7,\ x - y = 2$	add the equations: $3x = 9$	$(3, 1)$	$6 + 1 = 7$
$3x + 2y = 12,\ x + 4y = 10$	$(1) \times 2$ against (2): $5x = 14$	$(\frac{14}{5}, \frac{9}{5})$	$8.4 + 3.6 = 12$
$5x - 3y = 11,\ 2x + y = 5$	$(2) \times 3$ then add: $11x = 26$	$(\frac{26}{11}, \frac{3}{11})$	$\frac{130-9}{11} = 11$
$6x + 4y = 20,\ 9x + 6y = 30$	both reduce to $3x + 2y = 10$	a whole line	$ae - bd = 0$

ASIDE / THE HINT IN PROBLEM 4

The question asks what you notice after dividing by the GCDs, and the answer is that two equations have become one.

ASIDE / CHECK IN PYTHON

`Fraction(c*e-b*f,a*e-b*d)` gives `42/11` for `(3,2,16,2,5,19)`.

Convention. Equations are $ax + by = c$ and $dx + ey = f$. After the Year 9 Algebra: Path to Simultaneous Equations page.